

Power BI Interview questions

Q1. What is Power BI?

ANS- Power BI is a Data Visualization and Business Intelligence tool that converts data from different data sources to interactive dashboards and BI reports.

Q2. Why Power BI?

- Power BI is user friendly
- This is easy to connect from our data & it can connect multi data sources
- This is easy to use & share to next user
- This is compatible for all kind of business organizations, whether its small or big
- This is available on all kind of devises, like laptop, mobile & other machineries

Q3. Power Pivot?

ANS - Power Pivot mostly used to data modeling-

Data Modeling – when we got the data from client those are in tabular format, so we have to create relationship between tables using primary & secondary key.

Q- What is incremental refresh in Power BI?

ANS- Incremental refresh will allow you to refresh only newly added data in the existing data.

Q- Difference between Dashboard & Power BI Report?

Power BI: -

1. Power BI report is combination of multiple visuals & elements like- chart, text, values on that page.
2. Power BI report is created on PB Desktop.

Dashboard: -

- 1- Power BI dashboard is the business key value that can change the business profit and can be glanced at one screen.
- 2- For Power BI dashboard first of all we have to publish the report from desktop to Power BI server after that we can create Power BI dashboard.

Q- What is CORR function?

ANS - CORR is a correlation function that provides correlation between two distinct values.

Q- What is data alert or data driven alert?

ANS - When data refreshed Power BI looks for an alert, and if it reached the alert threshold or limit then the alert will be triggered.

Q- What is Power BI Desktop?

ANS - Power BI Desktop is a data analysis and reporting application user can install on a computer to create dashboards and live reports. Power BI desktop integrated with Power BI Service, the user can also share these reports with decision-makers and stakeholders to help them understand the current state of the business.

Q- What is Power Query?

ANS - Power Query is tool that filters, transform and combine the data from various sources, it helps to import the data from database.

Q- Types of filters.

ANS - there are filter based on conditions

- Visual level filter – This filter works on visual only
- Page level filter – This filter works on page only
- Report level filter – This filter works on all report

Q-What is difference between custom column & conditional column?

Conditional column- can't work on multi columns condition.

Custom column – can work on multi columns condition.

Q- What is KPI in Power BI?

ANS- KPI is abbreviation of Key Performance Indicator. Through the KPIs, organizations can evaluate their success at reaching the target.

Q- What is DAX?

ANS - Data Analysis Expression, used to generate new information (Data) by performing calculation on existing data.

Q- Deference between filter & Slicer.

FILTER - Restrict to user to view and access the data. In filter have

Multi filter option have like- greater than or less than

In filter, do not have this function

SLICER - User can interact with data via slicer

Slicer don't have

We can select none in edit interaction to avoid change the data.

Q- What are the various types of refresh?

1-Packaged refresh: - It synchronized Power BI Desktop or excel file between the **Power BI Services** and **one drive** or **share point**.

2-Model or Data refresh: - this refresh the data available in the Power BI Service with the data stored in original data sources.

3-Tile refresh: - this feature update the cache for Tiles at every 15 min on the dashboard.

4-Visual Contained refresh: - Used to refresh the visual containers, once data got changed.

Q- Can we have more than one functional relationship between two tables in a Power Pivot Data Model?

ANS- **No**, there can be multiple inactive relationship, but only one active relationship between two tables in a Power Pivot Data Model. **Dotted lines** represent **inactive relationship** and continuous lines represent **active relationship**

Q- Which function used to use inactive relationship?

ANS- **USERELATIONSHIP ()**, is used to take value from inactive relation, this relationship can be useful in specific scenarios.

Note- Before using this function create INACTIVE RELATION then use this function.

Q- What is Row level security, how to use Dynamic RLS?

ANS- Row level security is defining on Desktop & assign to services, under this only allowed user can view the data/report which is assigned to them.

For Example - One table has multiple regions, one table has multiple departments & multiple client data, in this scenario user can view the report which is relevant to him, for this we use Row Level Security.

There are two types of RSL

1. Static RLS
2. Dynamic RLS, mostly we use Dynamic RLS.

For dynamic we use USERPRINCIPALNAME function for creating the dynamic RLS.

What is dynamic RLS- if someone working at any particular location and moving to other location then there is no need to change anything, on the basis of his email ID he can access data of his location.

How to apply—**Go to modeling –Manage Roll –Create new roll – select --USERPRINCIPALNAME ()**

Q- How we can improve performance / Performance Tuning of reports

1. Remove unnecessary columns & tables where possible
2. We can create aggregations to improve the performance
3. Use Microsoft app store certified visuals
4. Use Import mode rather than Direct Query.
5. Use performance analyzer, by this tool we can analyze speed of our dashboard & we can printout bottlenecks
6. We can limit visuals on the report page / avoid many to many relationships

Q- What is DAX function optimization / How to improve the performance measures

- In this case we use summarize function with required column and do the calculation in a **virtual table**.
- Create the measures on required filter condition rather than doing via filter/slicer if possible

Q- Have you worked on connection of API?

ANS - I have little knowledge but not worked.

Q- What are the building block in power bi services?

ANS- 1. Dataset 2. Report 3. Dashboard 4. Tiles 5. Visualization.

Q- What is parameter?

ANS- Parameter is used for reusability; we can define parameter for different type of use, for example - to create start date & end Date parameter and parameter is also used to change the data source.

Q-What is Field Parameter / How will you show different calculation for each visual if you are using slicer.

Ans- With the help of fields Parameter, we can apply multi visuals on same time on same visual.

Q-How to show incremental value / What if analysis.

ANS- How to show – PBD—Modeling—New Parameter—Numeric range—Create parameter—Assign this in measure.

Q- Mean of Calculated tables / Calculated columns / Measures.

- **Calculated tables:** - Returns a table based on filter conditions. It returns all columns of the base table.
- **Calculate column:** - Where we require aggregate function with filter we use calculate
 - It will store new physical column
 - It will be part of Data Model
 - It will calculate during the loading or refresh
 - It will not change as per filter
- **Measures:** - Measures are used to perform the calculations on aggregated data. They calculate values based on the context, such as filters, slicers, and row or column context
 - It not stores in database; it executes at run time
 - It is not part of Data Model
 - It performance is better than calculated columns
 - This will get changed as per filter

Q- What is difference between Group by & Summarize DAX function?

Group By- you can group values, various rows into a single value by grouping the rows according to the values Group by used with CURRENTGROUP ().

Summarize: - SUMMARIZE DAX Function Creates a table of the input, grouped by the selected columns.

Q- Type of mode/connection mode in Power BI.

- 1- **Import mode-** we import sources to Power BI cache, we can import 1 GB data.
- 2- **Direct Query mode-** We create connection to data sources, we can import 10 lacs rows, in this way Data is not visible in table view, when we select any visual or filter that time created connection fire a SQL Query.

Limitations of Direct Query: -

1. if we create measures based on time function
2. If data changes, you must refresh to show the latest data
3. You must refresh in Power BI Desktop to reflect schema changes.
4. You can't change a mode from Import to Direct Query mode but Direct query to Import mode can be change

Note: - We can view table in Power Query, but in Power BI Desktop under “Table View” we can't not view the table.

Benefit of Direct Query: - Visualization can be on very large data set, real time reporting, Direct Query can be change to Import Mode

Benefit of Import Mode: - Import Mode provides better performance (data is already in cache), can use all DAX function, can combine multi tables imported from different sources.

Q- What is the live connection?

ANS- When we import data from Power BI Services by selecting the semantic model to create the visualization, in this way data view will not work only connection created.

Q- Difference between live connection & direct query?

- ❖ **Direct Query** is a direct connection to data sources, for example, SQL Server, Oracle, IBM.
- ❖ **Live connection** is a direct connection to the Semantic model (Azure AS, SSAS Tabular, Multi-Dimensional, or a Power BI report published in service)

Q- What is Mashup/M Language/ what are supporting function?

ANS- "M" is programming language used in Power Query, this is case sensitive language similar to other programming language. Syntax start from **let....in**, below are some syntax of M query

- **Excel.workbook**
- **Table.Combine**
- **Table.PromoteHeaders**
- **Table.TransformColumnTypes**

Q- What is alternative for Nested if in Power BI?

ANS- **SWITCH** This function works like **if function**.

Q- Difference between summarize & summarize columns?

- **Summarize** -This is same as Group by, but in this function blank will not be ignored.
- **Summarize columns** – This function also works same but with filter & will ignore blank & execute **faster than others**.

Q- Difference between Append & Merge.

- **Append**- this will paste data to a table at **last row** wise.
- **Merge**- this will merge the table **column wise** as per types of joins (left, right, full and inner join)

Q- What are the different types of join in power BI?

1. Horizontal join are used to merge data
2. Vertical join is used to append data

Q- What does mean of Fuzzy Matching Option?

ANS- under this join will happen with approx. matching ID's.

Q- Difference between Related & Related table DAX function?

- Related- can be used many to one relationship, under this only one column can be used
- Related Table- can be used one to many relationships, under this multi column can be used

Q- Is Many to one will work with related table?

ANS-No, here used RELATED function

(Dup data/Fact table) Many side related used

(Unique data/Dimension table) Single side- related table used

Q- Roles in Workspaces?

Ans- There are 4 types of role we can assign in Power BI Services

1. **Admin**- Update and delete the workspace
2. **Member**- Add or remove people, including other admins.
3. **Contributor**- Modify gateway connection settings
4. **Viewer**- View and read content of data

Q- How many types of schema are available in power BI?

- 1- **Star Schema** – Star schema dimension table directly connect to fact table
- 2- **Snowflake Schema** – under this dimension table firstly connect to another dimension then connect to fact table, so calculation is difficult **(we use snow flake schema to reduce the duplicity redundancy)**

Q-What is Gateway

ANS- This is a connector between- Report & data sources, this is required for refresh the report, there are two types of gateway:
1- On premises gateway/Standard Mode—Multi users can connect to the data sources, (48 times can refresh in a day)
2- Personal on premises gateway/Personal Mode—Only single user can connect to data sources (8 times can refresh in a day)

Gateway **not** required for below: -

- share point
- Azure SQL DB
- CRM Online
- For online or cloud-based data sources

Q- Automatic refresh

ANS- For this go to Workspace—click on 3 dot of SEMENTIC MODEL of report>> Setting>>Refresh>> **switch on of refresh schedule** and assign the **times** to get refresh (**Gateway is required for this**)

Q Benefit of variable

1. Improve performance & Readability
3. Simply debugging
4. Reduce Complexity

Q-What is edit interaction?

ANS- Format→When we click on any visual & filter all the visuals data gets filter, if we don't want to filter other visuals for that we use Edit interaction, there are 3 types of interaction in power PBI 1. Highlight 2. Filter. 3. None, by default is Filter

Q-What is time intelligence function.

ANS- Time intelligence functions are used to perform calculation over the time period, we can perform calculations Daily, monthly & yearly.

Q-What is cross filter?

ANS- Cross filter enables to filter on **both side** of the relationship between the tables, that solves many to many relationship problems. `CALCULATE (DISTINCTCOUNT(Customer[Customer Name]),CROSSFILTER(Orders[Cust-ID],Customer[Cust-ID],Both))`

Q-Most common data shaping techniques.

ANS- Removing column, Adding index, Sorting.

Q- What is embed code in power bi services

ANS- it is an option in power bi services, used to generate link address for power bi reports and can also share across clients

Q- What is the bridge table/why we create?

ANS- If we have to join two or more table, and both table have dup data, then we create **Bridge Table** to join all three tables and can get correct result. `=Distinct (union (values (city of 1table), Values (city of 2table)))`

Q- Difference between MAX () and MAXA ()?

- MAX- will return the highest value of column
- MAXX- will return the highest value after having the calculation by each row.

Q- Mean of Cardinality (प्रमुखता) and use of this?

ANS-Cardinality shows the way of joins

Once joins happen one to many or many to one.

Q- How we can get top 6 sale, all possible ans?

ANS- `CALCULATE([TotalSale], KEEPFILTERS(TOPN(5,ALL(DmiCalendar[Date]),[TotalSale],DESC)))`

Q- What is Top and Rank function?

Top- list of values to the Top N number of values, such as top 10 or top 25.

Rank- under this all values aligned by order & showing rank of all values.

Q- How many ways to share the report to end users, if report is published on services.

ANS- Work space access, App, Embed URL.

Q-Use of Development pipeline.

ANS- Deployment pipelines enable creators to **develop** and test Power BI content in the **Power BI service** before the content is made available for users. It offers creators improved productivity, faster delivery of contents, and reduced manual work and errors. The tool is designed with three stages: development, test, and production.

Q-How can you handle errors during data transformation in Power Query?

Remove errors

Replace errors

Keep errors

Data type conversion errors

Q- What is Query Folding? And its benefit?

ANS- A query statement written as text in the database as SQL query, this is used to retrieve & transform from data source, it can be viewed in Power Query by right clicking at gear Icon in query setting, there is option of View Native Query to check the query folding. It works in both **Direct Query mode & Import mode**.

Benefit- Improved query performance
Reduced data transfer
Optimized resource utilization

Q-What types of license we have & what license using in you company.

ANS- Power BI premium license using (because in pro- can't use some features, dev+pipeline not used).

1-Pro-> we can import 1 GB of data, 8 time Scheduled refresh are allowed in pro version, cost of pro licensing is 10\$ Per month

2-Premium-> we can import 10 GB Data, we can set 48-time schedule refresh, cost of premium is 20\$ per month & as per capacity

3-Fabric

Q- What is the capacity for premium use

1. Dedicated Resources
2. Enhanced Performance
3. Larger Data Capacities
4. Advanced Features
5. Additional Licensing Options

Q-How to calculate rolling sale /running sale/cumulative sale?

ANS- A running total is a sequence of partial sums of any given data set.

Calculate([TotalSale], Filter(All(DimCalendar),DimCalendar[Date]<=MAX(Dimcalendar[Date])))

Q- What is difference between Date add & SAMEPERIODLASTYEAR?

DATEADD- With this formula we can compare sale by DAY, MONTH, YEAR & QUARTER in positive & negative.

SAMEPERIODLASTYEAR- With this formula we can get only Last Year sale

Q- Difference In between Row context & Filter context.

- Aggregate function works as Column context/Filter context
- Iterator Function works as Row context

Q- What are the charts you have used in your project?

We have 30+ charts in visual pan, we have used Colum chart, Pie charts, Donut charts, KPI & Map, Water Fall chart, Cards...

I had a situation to show comparison year on year, I used water fall chart

I had situation to show marketing strategies I used Funnel Chart

& sometimes need to show key numbers I used to show in cards & I have used Multi card charts also.

Q- What custom charts you have used?

ANS- We can import charts by clicking on 3 dot at visual pan in Power BI Desktop, there are 250+ custom chart available such as- drill down charts there, infographic chart there Scroller charts are there.

Q- How can you make your report Dynamic?

ANS- By making use of Bookmarks, by selecting visuals & by making certain use of button & action we can make **report Dynamic**.

Q- Have you used Q&A visual?

ANS- This is available in Microsoft app store and can be used frequently, under this we can type our query for visual and this will show the same visual, and also recommend visual based on AI.

Q- How you will pull only number from column if you have data like Rs3000 & \$200.

ANS- Will use replace function for all symbol

Q- How to change snowflake schema to star schema/Why we use Snowflake schema?

ANS-

Q- If client requires start date is 1st Apr & end date is 31st Mar (financial year), how will you will implement.

ANS- FinancialYear = IF(DimCalendar[MonthNo]>=4,DimCalendar[MonthNo]-3,DimCalendar[MonthNo]+9)

Q- What challenges you have faced?

Since I have worked with few reports and some of them are on Direct Query connection mode, it becomes quit challenging

1. If we create measures based on time function
2. Issue related of color blindness from end users point of view (go with dark theme)
3. RLS issue, sometime client ask for security should be on 1st page for reason, but on 2nd page should be content of my region, so it is quit challenging.
4. To educate each & every end user about the functioning of report

Q- What KPI you have used for making the reports

ANS- Can explain.... we used Card to show the KPI as total Sale, Profit, average sale etc. and explain other charts also

Q-Power BI Components

1. Power Query
2. Power Pivot (modeling)
3. Power View (visualization pan)
4. Power Map (visualization pan)
5. Power Q&A
6. Power BI Desktop
7. Power BI Website
8. Power BI Mobile Apps

Q- Power BI Report Server

ANS-The Power BI Report Server is similar to the Power BI Service. The only difference between these two is that: - **Power BI Report Server is an on premise platform**. It is used by organizations who do not want to publish their reports on the cloud due to confidentiality & security of data.

Q- What is paginated Report?

ANS- Paginated reports are designed to be printed or shared. They're called paginated because they're formatted to fit on a page. if I print the report they display all the data in multiple pages.

Note- Paginated report can be deployed only to **premium capacity** workspace.

Report Builder requires for this report.

Q- What is difference between Duplicate & Reference?

Duplicate	Reference
A duplicate query behaves as an autonomous query	Reference query is directly linked to its parent query.
Any changes in the parent query does not impact the duplicate.	Any changes in the parent query directly impacts the referenced query.
Each step needs to be performed independently.	What steps performed in parent query will impact to referenced table

Q- What are the tools of Power BI?

1. Modelling Ribbon
2. Fields list
3. Navigation Pane
4. Visualization Pane

Q- What is mean of Data Set, Data Flow & Data Mart?

Data set- The dataset in Power BI is a source of data that you import, connect, and make visualizations It can be easily seen in the Power BI services. Power BI Dataset is used commonly when sharing a model between multiple visualization objects.

Data Flow- This is all about Self-service data preparation, whatever steps we perform in power query to transform data, same steps we can perform on dataflow to transform the data and we can use this data flow once we required for same process.

Data Mart- DataMart provides a complete ecosystem for **BI development**. Users can perform end-to-end data **preparation** & **exploration** without having coding experience. This ensures that developer and **non-technical** team can work on **Power BI** without the help of a database administration.

Q- What is bookmark

Ans A bookmark store the current state of the page. It includes the changes what I made to **filters**, **sliders**, and **visuals** on that page

Q What is sync slicer

Ans- If we have done any selection on one page & navigating to next page same selection should show is call sync slicer

Q-How to get time of refresh (when data was refreshed)?

Ans- Go to PQ-->Menu-Home-newSource--blankquery-->Type in formula bar--"**DateTime.FixedLocalNow()**"--> click on to table (to convert) at top--close apply & use in Card

Q- Why it is called semantic data model?

Ans- Power BI allows users to modify existing data models in the Power BI service, such as editing relationships, creating DAX measures and managing RLS. In this experience, users can work and collaborate on the same data model.

Q- Cross report drill through

Ans- This is used to toggle on one report to another report for detailed analysis for this there are two reports: -

1- parent report 2- Details Report we have to do check in **Option** for both report "Cross-Report Drill Through" and in **Details Report** switch on the "Cross Report".

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Q-What are the TEXT Transformation Tool in Power Query?

ANS- Split columns, Format, Extract

Q- What Does PIVOT & UNPIVOT do in Power Query?

PIVOT: - rotates column from vertical to horizontal.

UNPIVOT: - rotates column from horizontal to vertical.

Q- What is Date & Time specific tool Do in Power Query

ANS- Check in Power Query Tool: -

Q- How do we look up in Power Query

ANS- With the help of merge function we can lookup data

Q- What is Indexing/how Index column can be added?

ANS- Index column contain a list of sequential value, that can be used to identify to each unique row in a table.

Q- How you group or Aggregate in Power Query?

ANS- Grouping help here to view & analyze trained in the visual.

Q- What is conditional column

ANS- Conditional column allows to define new fields based on logical rule & conditions.

Q- What is Data Source Setting in Power Query

ANS- Data Source setting in Power Query allows to manage data connections & permissions.